AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on SSC-II Computer Science Annual Examinations 2024

This document provides structured marking schemes, 2025 syllabus alignments, examiner commentary, and complete model solutions (in both C and 2025 JavaScript) for the AKU-EB SSC-II (Class X) Computer Science Annual Examinations 2024 Paper II.

General Observations: Candidates demonstrated solid understanding of flowchart symbol conventions, loop trace tables, and online store cybersecurity practices. Teachers should reinforce: (1) Using the relational equality operator (`==`) rather than assignment (`=`) in `if` conditions; (2) Using the modulus operator (`%`) for divisibility checks such as leap year verification; and (3) Variable shifting inside `for` loops for sequence generation such as the Fibonacci series.

Q.1 (2 Marks) — Original SLO: 7.3.3 | 2025 SLO: 9.1.2

Checking Hints: - 0.5 mark for each correctly matched/drawn flowchart symbol (4 × 0.5 = 2 Marks).

Better Responses: In better responses, candidates accurately drew or identified the Rectangle for A (calculations/processing), the Flowline Arrow for B (connecting symbols), the Parallelogram for C (input/output), and the Circle for D (on-page connector).

Weaker Responses: In weaker responses, candidates swapped the flowline arrow (B) and on-page circle connector (D), or wrote symbol names without drawing/identifying the corresponding shapes from the table.

Q.2 (3 Marks) — Original SLO: 10.1.8 | 2025 SLO: 10.4.3

Checking Hints: - 1 mark for calculating the sum of the three angles (`sum = a1 + a2 + a3;`).
- 1 mark for writing the `if` condition checking whether `sum == 180` (and printing that the triangle is valid).
- 1 mark for writing the `else` branch printing that the formation of the triangle is not possible.

Better Responses: In better responses, candidates computed `sum = a1 + a2 + a3;`, used the relational equality operator `if (sum == 180)` (often additionally verifying positive angles), and printed clear messages in both the `if` and `else` blocks.

Weaker Responses: In weaker responses, candidates wrote `if (sum = 180)` (which always evaluates to true in C), omitted the `sum = a1 + a2 + a3;` calculation, or used `scanf` instead of `printf` for output.

Q.3 (3 Marks) — Original SLO: 11.1.8 | 2025 SLO: 10.5.4

Checking Hints: - 1 mark for correctly completing Iteration 1 (`x = 14, y = 20, Output = 14`).
- 1 mark for correctly completing Iteration 2 (`x = 16, y = 20, Output = 16`).
- 1 mark for correctly completing Iteration 3 (`x = 18, y = 20, Output = 18`) and showing `x = 20, y = 20` where the loop terminates.

Better Responses: In better responses, candidates respected the statement order inside the `while` loop (`printf` first, then `x += 2`), recording outputs `14`, `16`, and `18` while keeping `y = 20` constant, and stopping when `x` reached `20`.

Weaker Responses: In weaker responses, candidates either incremented `x` before printing (producing `16, 18, 20`) or modified `y` across rows.

Q.4 (5 Marks) — Original SLO: 12.2.7 and 12.2.6 | 2025 SLO: Out of 2025 Class X Syllabus

Checking Hints: - **Part i (3 Marks):** 1 mark for NOT gates on `A` and `B` ($\overline{A}, \overline{B}$); 1 mark for the three OR gates representing $(A + \overline{B})$, $(\overline{A} + \overline{B})$, and $(A + B)$ (or equivalent Sum-of-Products gates); 1 mark for combining them with an AND gate to output X.
- **Part ii (2 Marks):** 0.5 mark for each correct row of the 4-row truth table (including intermediate working columns).

Better Responses: In better responses, candidates represented 'sends value 0' using NOT gates, connected the three two-input gates into a final combining gate, and showed complete intermediate columns in the Working Space of the truth table.

Weaker Responses: In weaker responses, candidates ignored the 'sends value 0' inversions (treating them as direct inputs A and B without NOT gates) or filled column X without showing intermediate steps in the Working Space.

Q.5a (6 Marks) — Original SLO: 10.1.8 | 2025 SLO: 10.4.3

Checking Hints: 1 mark for variable declaration; 1 mark for taking input; 2 marks for leap year condition (`year % 4 == 0`); 2 marks for printing leap year / not a leap year output.

Better Responses: In better responses for Part (a), candidates used the modulus operator `%` to test divisibility by 4 (and 100/400) inside an `if-else` statement. For Part (b), better responses initialised `t1 = 0` and `t2 = 1`, ran a `for` loop 10 times, printed `t1`, and updated `nextTerm = t1 + t2; t1 = t2; t2 = nextTerm;`.

Weaker Responses: In weaker responses for Part (a), candidates wrote `year / 4 == 0` instead of `year % 4 == 0`. In Part (b), weaker responses hard-coded `printf("0 1 1 2 3 5 8 13 21 34")` without computing terms dynamically or failed to shift `t1 = t2` and `t2 = nextTerm` in the correct order.

Q.5b (6 Marks) — Original SLO: 11.1.5 | 2025 SLO: 10.5.3

Checking Hints: 1 mark for initialising `t1 = 0, t2 = 1`; 2 marks for `for` loop header (`1` to `10`); 2 marks for computing `nextTerm = t1 + t2` and shifting variables; 1 mark for printing output.

Better Responses: In better responses for Part (a), candidates used the modulus operator `%` to test divisibility by 4 (and 100/400) inside an `if-else` statement. For Part (b), better responses initialised `t1 = 0` and `t2 = 1`, ran a `for` loop 10 times, printed `t1`, and updated `nextTerm = t1 + t2; t1 = t2; t2 = nextTerm;`.

Weaker Responses: In weaker responses for Part (a), candidates wrote `year / 4 == 0` instead of `year % 4 == 0`. In Part (b), weaker responses hard-coded `printf("0 1 1 2 3 5 8 13 21 34")` without computing terms dynamically or failed to shift `t1 = t2` and `t2 = nextTerm` in the correct order.

Q.6a (6 Marks) — Original SLO: 14.1.1 | 2025 SLO: 13.1.2

Checking Hints: i. 1 mark for each valid point on why computer security is vital (3 Marks).
ii. 1 mark for each specific security measure/practice (3 Marks).

Better Responses: In better responses for Part (a), candidates separated the business importance points (i) from the concrete technical safeguards (ii). For Part (b), candidates defined a virus as host-dependent self-replicating malware and explained four distinct transmission vectors.

Weaker Responses: In weaker responses for Part (a), candidates repeated the same point (e.g., 'install antivirus') in both parts i and ii. In Part (b), some candidates defined a virus merely as 'something that slows down the computer' without mentioning malicious code or replication.

Q.6b (6 Marks) — Original SLO: 14.1.4 | 2025 SLO: Out of 2025 Class X Syllabus

Checking Hints: i. 2 marks for describing a computer virus.
ii. 1 mark each for any FOUR ways a virus spreads (USB drives, email attachments, pirated software, networks/internet).

Better Responses: In better responses for Part (a), candidates separated the business importance points (i) from the concrete technical safeguards (ii). For Part (b), candidates defined a virus as host-dependent self-replicating malware and explained four distinct transmission vectors.

Weaker Responses: In weaker responses for Part (a), candidates repeated the same point (e.g., 'install antivirus') in both parts i and ii. In Part (b), some candidates defined a virus merely as 'something that slows down the computer' without mentioning malicious code or replication.